

Federated Learning With a Temporal-Degradation-Aware Transformer for Robust Joint SOC and SOH Estimation in Electric Vehicle Batteries

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Abstract—Accurate estimation of state of charge (SOC) and state of health (SOH) is critical for the safe and reliable operation of electric vehicle (EV) batteries. However, centralized approaches face challenges such as data privacy, communication overhead, and poor scalability under nonindependent and identically distributed (non-IID) data. Although some methods attempt joint estimation, most rely on centralized learning, lack computational efficiency for embedded systems, and overlook uncertainty quantification (UQ), limiting real-world applicability. To address these limitations, this article proposes a federated framework for joint SOC and SOH estimation that is privacy-preserving, lightweight, and uncertainty-aware. The core model is a temporal-degradation-aware informer (TDA-Informer), which incorporates an Informer-Lite encoder and a multitask prediction structure. A battery-focused multifeature attention (BFMFA) module emphasizes critical inputs such as voltage, current, and temperature. The stochastic controlled averaging for federated learning (SCAFFOLD) algorithm is employed to mitigate client drift under non-IID conditions, enhancing convergence across distributed EV clients. UQ is achieved via Monte Carlo Dropout to provide confidence intervals for SOC and SOH predictions. The framework is trained and validated on the NASA battery dataset and tested for generalization on the unseen Oxford dataset, with realistic sensor noise added to simulate practical conditions. Extensive experiments, including ablation studies and comparisons with baseline models, confirm its superior accuracy, robustness, and scalability. While physical deployment remains future work, the results confirm the framework's potential for intelligent, scalable battery management systems (BMSs).

Index Terms—Battery management system (BMS), electric vehicles (EVs), federated learning (FL), state of charge (SOC), state of health (SOH), Transformer network, uncertainty quantification (UQ).

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I. INTRODUCTION

THE global rise of electric vehicles (EVs) has increased the demand for reliable and long-lasting lithium-ion batteries, which serve as the primary energy source for EV propulsion. Accurate estimation of key internal states, specifically state of charge (SOC) and state of health (SOH), is essential for safe operation. SOC indicates the available energy and directly affects the driving range, whereas SOH reflects battery aging and capacity loss, which are critical for predicting remaining useful life (RUL) and scheduling maintenance [1]. However, accurate estimation remains challenging due to the complex electrochemical behavior of batteries, high-rate operating conditions [2], and limited sensor observability. Deep learning (DL)-based methods have shown promise, yet most rely on centralized training, where data from multiple vehicles is aggregated on a server [3], [4]. This raises two major issues for large-scale deployment: privacy and data heterogeneity. Battery usage data may expose sensitive patterns, limiting data sharing due to privacy regulations and proprietary concerns [5], [6], [7]. Moreover, EV battery data is highly nonindependent and identically distributed (non-IID), influenced by differences in battery types, user behavior, charging patterns, and environmental conditions [8], [9]. These factors underscore the need for a privacy-preserving, generalizable framework for joint SOC and SOH estimation from distributed and diverse real-world battery data.

A. Literature Review

Battery state estimation has seen rapid advancements driven by data-driven techniques. Early approaches successfully employed models such as convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and gated recurrent units (GRUs) for either SOC estimation [10], [11] or SOH prediction [12]. Several studies focused on extracting health indicators from charging profiles and tuning traditional machine learning models like extreme learning machines (ELMs) using optimization methods such as particle swarm optimization (PSO) [1], [13]. To enhance robustness and adaptability, hybrid architectures combining CNNs and LSTMs [14] and transfer learning (TL) techniques for model adaptation across domains have also been proposed [3], [15]. Furthermore, recent research has increasingly emphasized sophisticated data-driven strategies. One such approach is model-data

fusion, which integrates physics-based models with data-driven methods like Vision Transformers [16], while others use hybrid deep neural networks with dimensional attention mechanisms to prioritize key features [17]. Advanced optimization algorithms have also been combined with multitask learning frameworks to improve SOH estimation accuracy [18]. In a similar vein, specialized TL frameworks have been developed to boost model adaptability under specific, challenging conditions. For instance, Chen et al. [19] proposed an adaptive transfer Enformer framework for managing power battery health at low temperatures, demonstrating how TL can mitigate performance degradation in targeted operational contexts. However, these models are typically developed under centralized learning paradigms, which limit their deployment in distributed and privacy-sensitive environments like EV fleets. However, these models are typically developed under centralized learning paradigms, which limit their deployment in distributed and privacy-sensitive environments like EV fleets.

Transformer-based models have recently gained prominence due to their ability to model long-range dependencies in time-series data [20], [21], [22], [23], [24]. Despite their accuracy, their computational complexity poses challenges for edge deployment in resource-constrained systems such as battery management systems (BMSs) [8]. To address these limitations, lightweight and partial-data learning models have been explored [25], [26]. While such methods improve efficiency, they often lack generalization across diverse operating conditions and typically omit uncertainty modeling, which is critical for real-world reliability. Uncertainty quantification (UQ) is essential for safety-critical applications like BMSs, where prediction confidence must be explicitly modeled. In centralized frameworks, several techniques have been proposed to quantify uncertainty, including Gaussian process regression (GPR) [27], ensemble-based methods [28], and hybrid fusion models [29]. These approaches enable the estimation of prediction intervals for battery states, thereby supporting risk-aware decision-making. However, they are often computationally intensive and not directly applicable to decentralized or latency-sensitive environments. Recent review studies have highlighted the need for solutions that integrate lightweight architectures with well-calibrated uncertainty estimation, especially under real-world constraints such as non-IID data and privacy preservation [30], [31], [32].

To address data scarcity and domain shifts, TL has been effectively applied to SOH estimation. These approaches typically involve pretraining a model on a data-rich source domain (e.g., one battery type) and then fine-tuning it with a small amount of data from a target domain (a new battery type), as demonstrated with LSTM-based frameworks and adaptive Gaussian processes [41], [42]. While powerful, this paradigm often relies on a centralized source dataset and a sequential, one-to-one transfer of knowledge. In contrast, this work utilizes federated learning (FL), which offers several distinct advantages for a large, distributed fleet of EVs. Rather than a sequential transfer, FL enables parallel and collaborative training of a single global model directly on user devices. This approach is privacy-preserving by design, as raw sensor data never leaves the vehicle. Furthermore, it is highly scalable and

facilitates continuous, multidirectional learning, where every vehicle in the fleet both contributes to and benefits from the collective intelligence. This makes FL uniquely suited for building a robust, generalized model that is continuously updated from real-world, heterogeneous data, which is the primary focus of the proposed framework. As a result, developing models that are both efficient and capable of producing uncertainty-aware predictions in federated settings remains an open research challenge.

To address data privacy and scalability, FL has emerged as a viable alternative to centralized training in battery prognostics [43]. FL enables distributed clients (e.g., EVs) to collaboratively train a global model without sharing raw data and has been applied to areas like EV charge scheduling and resource optimization [35], [44]. Early FL applications for SOC and SOH estimation adopted the basic federated averaging (FedAvg) algorithm [5], [7], [34], but FedAvg performs poorly under non-IID data, common in heterogeneous EV fleets [9]. This challenge, known as “client drift,” occurs when local updates diverge due to data heterogeneity, hindering convergence and generalization [8]. To overcome these limitations, a new generation of advanced FL algorithms has been proposed. Some approaches introduce contribution-aware aggregation methods that assign weights to client updates based on data quality or model performance [36], [37]. Building on this, other works have explored value-informed federated strategies to optimize the aggregation process. For example, Chen et al. [38] developed a global-local attention network combined with a value-informed federated strategy, which assigns aggregation weights based on the contribution of each client’s update to the global model’s performance, thereby improving SOH prediction accuracy. While these methods enhance aggregation, others improve alignment by matching functionally similar neurons before aggregation [8]. FedProx addresses client drift by adding a proximal term to the local objective function [45], whereas stochastic controlled averaging for FL (SCAFFOLD) employs control variates to correct gradient discrepancies between clients and the server during training [46]. More sophisticated frameworks, such as personalized federated TL (PFTL), aim to tailor global models for individual clients [39]. Similarly, recent industrial large model frameworks have explored the use of domain knowledge and lightweight “adapter” modules that are fine-tuned on target-specific data to achieve high-fidelity personalization [47]. Source-free FL approaches further enhance privacy by transmitting only statistical summaries rather than raw model updates [6]. These developments indicate a clear shift from centralized, generic models to distributed, personalized, and privacy-preserving architectures. To provide a structured overview of this evolution, a comparative summary of existing methods is presented in Table I, highlighting the growing emphasis on generalization, uncertainty modeling, and privacy within the state-of-the-art battery estimation landscape.

B. Research Gap and Contributions

1) *Research Gap:* Despite notable progress in battery state estimation, a unified and practical framework suitable for

TABLE I
COMPARATIVE ANALYSIS OF STATE-OF-THE-ART METHODS FOR BATTERY STATE ESTIMATION

Ref. & Category	Type	Model Architecture	Key Feature	Federated Strategy	Reliability/Privacy	Generalization
Centralized Learning Methods						
[1], [12], [13]	SOH	GRU-CNN / ELM / etc.	Feature Engineering	N/A (Centralized)	None	Low
[3], [4]	SOC+SOH	LSTM / GRU + Transfer Learning	Domain Adaptation	N/A (Centralized)	None	Moderate
[19]	SOH	Adaptive Transfer Enformer	Low-Temperature Adaptation	N/A (Centralized)	None	Moderate
[20]–[24], [33]	SOC/SOC+SOH	Transformer Variants	Attention Mechanism	N/A (Centralized)	None	Moderate
[27]–[29]	SOH/RUL	Data-Driven Fusion (GPR, etc.)	UQ	N/A (Centralized)	UQ	Moderate
Federated Learning Methods						
[7], [34]	SOH	1D CNN / LSTM	Basic FL Implementation	FedAvg	None	Low-to-Moderate
[5], [35]	SOH/SOC	LSTM / etc.	EV Fleet Application	FedAvg / FedACM	None	Moderate
[9], [36], [37]	SOC/SOH	LSTM / AFT-Conv2d	Non-IID Data Handling	Proximal Term / Weighted Avg	None	Moderate-to-High
[38]	SOH	Global-Local Attention Network	Value-Informed Strategy	Value-Aware Aggregation	None	High
[6], [8], [39], [40]	SOH / RUL	LSTM / Custom	Advanced Privacy / Personalization	FedMA / PFTL / Source-Free	Differential Privacy	High
This Work	SOC+SOH	Lightweight TDA-Informer	Joint Estimation, Edge-Ready	SCAFFOLD	UQ	High

Note: Generalization reflects robustness across varied datasets and non-IID conditions. PFTL is Personalized Federated Transfer Learning.

real-world EV fleets remains lacking. Most existing works focus on isolated challenges, with centralized models limited by privacy concerns [3], [4], and early FL approaches relying on FedAvg, which performs poorly under the non-IID conditions typical of EV data [5], [7], [8], [9]. While recent FL methods address data heterogeneity and personalization [36], [39], they have not been combined into a comprehensive solution that simultaneously offers: 1) a lightweight model for joint SOC and SOH estimation; 2) a robust FL strategy, such as SCAFFOLD, to mitigate client drift; and 3) integrated UQ for reliability. The absence of such an efficient, generalizable, and uncertainty-aware framework limits the real-world deployment of intelligent BMS.

2) *Contributions*: To address the above research gaps, this work presents the following key contributions.

- 1) We propose a temporal-degradation-aware Transformer (TDA-Informer) with a lightweight Informer-Lite encoder. The model reduces computational complexity while jointly estimating SOC and SOH through a multitask structure. Although tested on public datasets, its efficiency suggests suitability for future deployment in resource-constrained BMSs.
- 2) To enhance both performance and interpretability, we introduce a battery-focused multifeature attention (BFMFA) module that prioritizes key inputs such as voltage, current, and temperature. The model is integrated into an FL framework using the SCAFFOLD algorithm, which effectively addresses client drift and supports joint SOC and SOH estimation under non-IID conditions commonly observed in EV fleets.
- 3) We incorporate UQ via Monte Carlo Dropout to provide confidence intervals for SOC and SOH predictions. Robustness is further validated through sensor noise

injection, demonstrating the framework's reliability for safety-critical applications.

- 4) The proposed framework is extensively validated through simulation studies, including joint SOC and SOH prediction accuracy, comparative analysis against baseline models, ablation experiments with different FL algorithms, and cross-dataset evaluation on unseen battery data to demonstrate its generalization capability.

The rest of the article is organized as follows. Section II describes the proposed methodology. Section III presents the simulation results and validation. Section IV concludes the article.

II. PROPOSED METHODOLOGY

The proposed framework integrates a lightweight, attention-based DL model within an advanced FL architecture to achieve robust, privacy-preserving, and computationally efficient joint estimation of SOC and SOH. The end-to-end pipeline, illustrated in Fig. 1, consists of three core components: 1) a lightweight TDA-Informer model that processes local time-series data; 2) an FL framework to collaboratively train a global model while addressing data heterogeneity; and 3) a UQ module to provide confidence measures for the predictions.

A. Input Data and Feature Engineering

Accurate joint estimation of SOC and SOH depends critically on the richness and diversity of the input data. In this work, we utilize three distinct battery datasets to ensure a comprehensive evaluation of the proposed framework. The primary dataset for initial training and validation is the widely used NASA Li-ion Battery Aging Dataset [48]. For generalization assessment, we employ two distinct strategies.

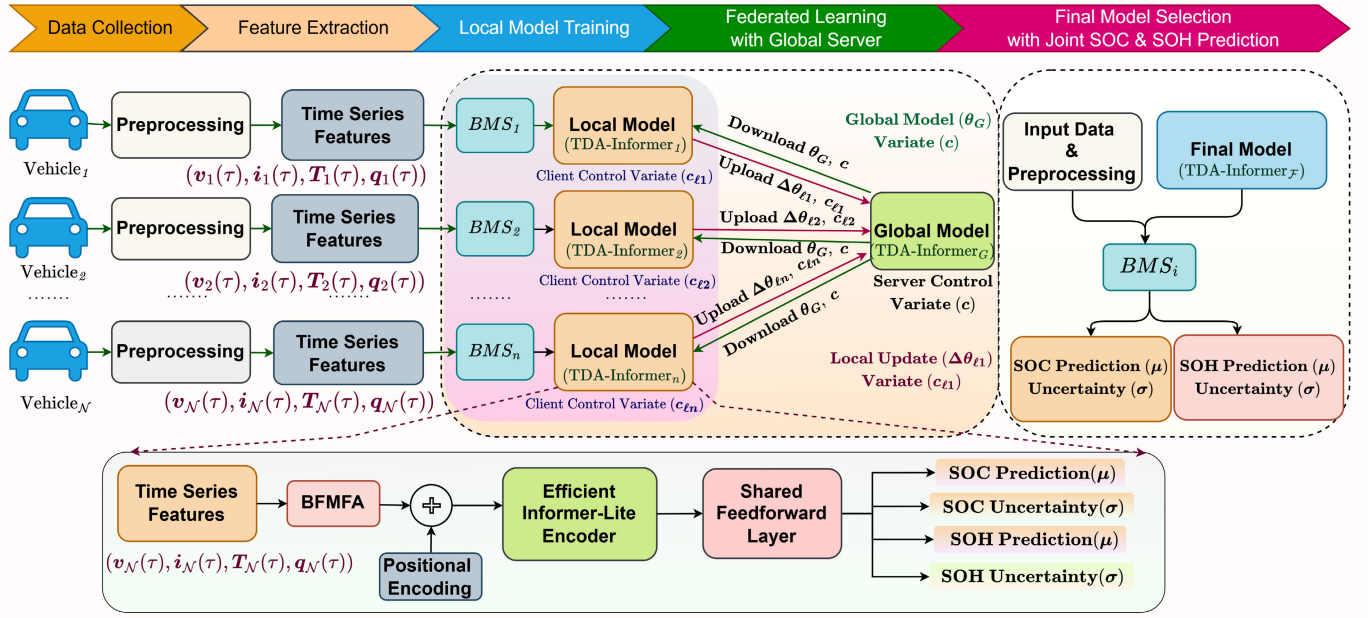


Fig. 1. Proposed federated TDA-Informer architecture for joint SOC and SOH prediction.

TABLE II
SUMMARY OF DATASETS USED IN THIS STUDY

Dataset	Source	Chemistry	Clients	Specifications	Primary use in this study
NASA	Saha & Goebel [48]	Li-ion (LCO)	22	2.0 Ah 2.7–4.2 V	Base model training and validation on aging data.
Oxford	Birkl [49]	Li-ion (LFP)	8	2.0 Ah 2.0–3.6 V	Generalization testing on an unseen dataset and chemistry.
Tongji	Zhu et al. [50]	NCA	9	3.5 Ah 2.65–4.2 V	Added to the training pool to enrich the model with modern chemistry data.
		NCM NCA+NCM blend	24	2.5–3.5 Ah 2.5–4.2 V	Used as two distinct, unseen test sets for robust generalization validation.

First, we test the trained model on the entirely unseen Oxford Battery Degradation Dataset [49]. Second, to further validate our approach on a larger and more modern dataset, we incorporate the comprehensive dataset from Zhu et al. [50], named the Tongji dataset. We partition this dataset by chemistry. The lithium nickel cobalt aluminum oxide (NCA, $\text{LiNi}_{0.86}\text{Co}_{0.11}\text{Al}_{0.03}\text{O}_2$) cells are included in the training data pool, while the lithium nickel cobalt manganese oxide (NCM, $\text{LiNi}_{0.83}\text{Co}_{0.11}\text{Mn}_{0.07}\text{O}_2$) cells and the NCA+NCM blend cells (a mixture of $\text{Li}(\text{NiCoMn})\text{O}_2$ and $\text{Li}(\text{NiCoAl})\text{O}_2$) are reserved as two separate, unseen test sets. This rigorous methodology allows us to evaluate the model's generalization performance not only across different datasets but also across different modern commercial chemistries. A detailed summary of these datasets is provided in Table II. Each dataset contains core battery operating signals captured at discrete time intervals τ , including terminal voltage $v(\tau)$, load current $i(\tau)$, cell or ambient temperature $T(\tau)$, and charge throughput $q(\tau)$. To optimize for the dual-task nature of the problem, we use

both instantaneous and cumulative features. The raw signals $v(\tau)$, $i(\tau)$, $T(\tau)$ are crucial for capturing the high-frequency dynamics needed for SOC estimation. Simultaneously, the charge throughput $q(\tau)$, which represents the cumulative charge passed through the battery, serves as a strong indicator of the total stress and is fundamentally linked to long-term SOH degradation. The combined feature vector for a battery over a time window of length Γ is therefore represented as

$$\chi = [v(\tau), i(\tau), T(\tau), q(\tau)] \in \mathbb{R}^{\Gamma \times 4}. \quad (1)$$

The time stamp information is implicitly captured by the sequence order and handled by the model's positional encoding. This rich feature set provides the model with the necessary information to learn patterns for both short- and long-term state estimation.

B. TDA-Informer Architecture

To address the need for a model that is both powerful enough to capture complex battery dynamics and efficient

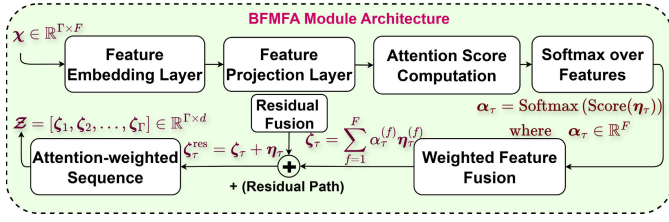


Fig. 2. Internal architecture of the BFMFA module. The module processes the input features χ by first projecting them into an embedding space η . It then computes attention scores (α_τ) to weigh the importance of each feature at a given time step. These weights are used to create a weighted feature fusion, which is combined with the original embedding via a residual path to produce the final attention-weighted sequence $\mathcal{Z}$.

enough for edge deployment on a BMS, we propose the TDA-Informer. This architecture features a shared, computationally efficient encoder that learns a common representation of battery behavior, which is then fed into two task-specific prediction heads for SOC and SOH estimation.

1) *BFMFA Module*: Before temporal feature extraction, the input sequence χ is processed by the BFMFA module (Fig. 2). This module acts as a feature-specific filter, dynamically learning the importance of voltage, current, and temperature at each time step. The BFMFA block projects each feature dimension into a latent representation

$$\eta_\tau^{(f)} = \mathbf{W}_f \cdot \chi_\tau^{(f)} + \mathbf{b}_f \quad (2)$$

where $\chi_\tau^{(f)}$ is the f th feature at time τ and $\mathbf{W}_f, \mathbf{b}_f$ are trainable weights. An attention score is then computed to quantify the importance of each feature

$$\varepsilon_\tau^{(f)} = \mathbf{a}^\top \tanh(\mathbf{U}\eta_\tau^{(f)} + \mathbf{c}). \quad (3)$$

The scores are normalized using a Softmax function, allowing the model to allocate differential weights to features dynamically

$$\alpha_\tau^{(f)} = \frac{\exp(\varepsilon_\tau^{(f)})}{\sum_{g=1}^F \exp(\varepsilon_\tau^{(g)})}. \quad (4)$$

The final attention-weighted feature embedding is obtained by fusing context from all features, enhancing the ability of the encoder to extract degradation-aware patterns

$$\zeta_\tau = \sum_{f=1}^F \alpha_\tau^{(f)} \eta_\tau^{(f)}. \quad (5)$$

The resulting sequence of embeddings $\mathcal{Z} = [\zeta_1, \dots, \zeta_\Gamma]$ is then combined with positional encoding. This specific ordering is intentional: the BFMFA module first creates a feature-rich representation for each time step, and the positional encoding is subsequently added to provide the necessary temporal context. This ensures that the model first determines *what* is important within each time step before the Transformer encoder considers *where* that time step occurs in the sequence.

It is essential to distinguish the proposed BFMFA module from the traditional self-attention mechanism found in standard Transformers. Conventional self-attention operates along the temporal dimension, calculating attention scores between

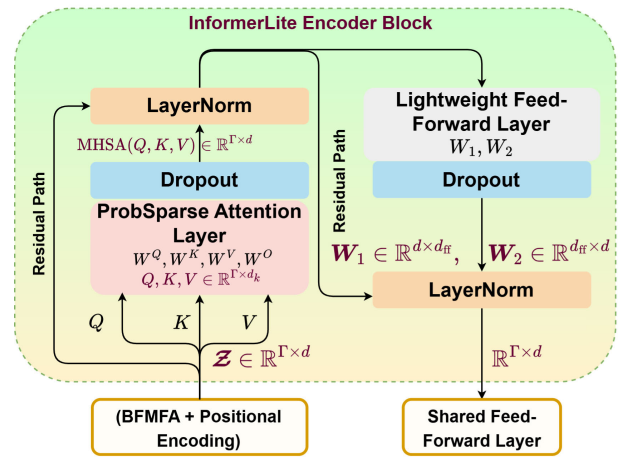


Fig. 3. Internal structure of the Informer-Lite encoder block.

different time steps (or tokens) in a sequence to model long-range dependencies. In stark contrast, the BFMFA operates exclusively on the feature dimension. For each individual time step τ , it computes attention weights *across* the various input features (e.g., voltage, current, and temperature) to determine their relative importance at that specific moment. Therefore, BFMFA acts as a dynamic, feature-wise filter that enhances the input representation *before* the main encoder models the temporal relationships, whereas traditional attention is designed to model those temporal relationships directly.

A key advantage of the BFMFA module's design is its inherent contribution to model interpretability. By generating explicit attention weights ($\alpha_\tau^{(f)}$) for each input feature (voltage, current, etc.) at every time step, the module provides a transparent mechanism to understand the model's focus. After training, these weights can be analyzed to determine which physical parameters the model deems most influential for predicting SOC and SOH under various operating conditions. For example, one could verify if the model correctly assigns higher importance to voltage during a standard discharge phase. This ability to qualitatively validate the model's internal logic against established physical principles moves the framework beyond a "black-box" paradigm, making it more trustworthy for safety-critical applications like a BMS.

2) *Efficient Transformer Encoder for Edge Deployment*: A standard Transformer encoder is often too resource-intensive for BMS hardware due to the quadratic complexity ($O(\Gamma^2)$) of its self-attention mechanism. To overcome this, our architecture's encoder is a custom, streamlined Informer-Lite implementation. It is specifically designed for edge efficiency by adapting the core principles of the well-established Informer model [51], particularly its ProbSparse self-attention mechanism. This mechanism achieves a significant reduction in complexity to $O(\Gamma \log \Gamma)$ by allowing each key to only attend to the most dominant "heavy-hitter" queries. By adopting this principle, our TDA-Informer is made suitable for real-time inference on resource-constrained edge devices. The encoder (Fig. 3) processes the sequence $\mathcal{Z}$ to effectively capture long-range temporal dependencies critical for modeling both SOH degradation and SOC dynamics.

3) *Joint Estimation via Multitask Heads*: The shared encoder produces a rich representation of the battery's state over time. This representation is then fed into two separate, fully connected prediction heads as follows.

- 1) *SOC Head*: An MLP that outputs the instantaneous SOC value for each time step in the input sequence.
- 2) *SOH Head*: An MLP that outputs a single SOH value representing the health of the battery for the entire input cycle.

This multitask learning approach allows the model to leverage shared patterns between the two related tasks, improving data efficiency and generalization, while the specialized heads ensure task-specific accuracy. A key consideration in this multitask design is the potential for negative transfer, where learning two distinct tasks could interfere with one another. The proposed framework mitigates this in two ways. First, the SOC and SOH tasks are synergistically related; a battery's long-term health directly governs its short-term charge/discharge dynamics. By learning them jointly, the shared encoder is compelled to develop a more robust and physically grounded representation of the battery's overall state. Second, while the encoder is shared to learn common features, the task-specific prediction heads are separate. This architectural choice allows for specialization, enabling the SOC head to focus on transient features while the SOH head focuses on long-term trends within the shared representation, thus preventing task interference at the prediction stage. The training process for these two tasks is balanced via the loss function. The total loss (L_{total}) is an unweighted sum of the mean absolute error (MAE) from the SOC prediction head (L_{SOC}) and the MAE from the SOH prediction head (L_{SOH}), formulated as $L_{\text{total}} = L_{\text{SOC}} + L_{\text{SOH}}$. This approach gives equal importance to both tasks, reflecting their equivalent criticality in a real-world BMS, and relies on the model's architecture to effectively manage the joint learning process.

C. FL for Heterogeneous Data

To enable privacy-preserving, collaborative training across a diverse fleet of EVs, we deploy the TDA-Informer within an FL framework. A critical challenge in real-world FL is the non-IID nature of data, which causes local client models to drift away from the global objective, a known issue that severely degrades the performance of the standard FedAvg algorithm. To address this, we employ the SCAFFOLD algorithm [46]. It is a state-of-the-art, communication-efficient algorithm that corrects for client drift. It achieves this by maintaining control variates on both the server (c) and each client (c_ℓ). These control variates track the update direction of the global model and the local models, respectively. During local training, the update rule for a client's model parameters θ_ℓ is modified to include a correction term

$$\theta_\ell^{(t+1)} \leftarrow \theta_\ell^{(t)} - \eta \left(g_\ell(\theta_\ell^{(t)}) - c_\ell^{(t)} + c^{(t)} \right) \quad (6)$$

where η is the learning rate and $g_\ell(\cdot)$ is the local gradient. This correction term ensures that the local updates are aligned with the global optimization trajectory, preventing drift and leading to faster, more stable convergence and a significantly

more accurate global model, especially in the presence of high data heterogeneity.

D. UQ via MC Dropout

For a safety-critical system like a BMS, knowing a model's confidence in its predictions is as important as the predictions themselves. To achieve this, we incorporate UQ into our framework using MC Dropout. During the inference phase, the dropout layers within the TDA-Informer architecture, which are typically deactivated, are kept active. A single input sequence is passed through the network K times. Due to the stochastic nature of dropout, this process generates a distribution of K slightly different output predictions for both SOC and SOH. The final prediction is taken as the mean of this distribution, which is a more robust estimate than a single deterministic pass. Crucially, the standard deviation of the distribution serves as a direct measure of the model's uncertainty. A high standard deviation indicates that the model is uncertain about its prediction, allowing the BMS to identify unreliable estimates and potentially fall back on safer operational parameters.

E. Physical Insights and Model Interpretability

Beyond the numerical validation, it is pertinent to analytically link the proposed architecture to the physical phenomena of battery degradation. The "temporal-degradation-aware" design reflects the architecture's suitability for battery dynamics, which consist of two superimposed timescales: fast-moving dynamics related to SOC (e.g., voltage response within a single cycle) and slow-moving dynamics related to SOH (e.g., gradual capacity fade and internal resistance increase over hundreds of cycles). These SOH changes are driven by electrochemical processes such as solid electrolyte interphase (SEI) layer growth, loss of lithium inventory (LLI), and loss of active material (LAM). The TDA-Informer's long-sequence forecasting capability is specifically structured to capture these long-term, slow-moving temporal dependencies. It can identify the subtle evolution in the shape of voltage/current curves over many cycles—the signature of degradation—rather than just the state within a single cycle. The BFMFA module provides a more direct link to physical processes. In a physical battery, V , I , and T are not independent factors; their interplay governs degradation.

- 1) Temperature (T) is a key accelerator for degradation, as higher temperatures exponentially accelerate parasitic reactions like SEI growth.
- 2) Current (I), particularly high C -rates, is a primary stressor linked to phenomena like lithium plating (especially at low temperatures) and mechanical stress.

The BFMFA's function, by operating on the feature dimension, is to learn a dynamic, nonlinear attention map for these inputs. This allows the model to learn, for example, that a high current (I) at a low temperature (T) is far more significant for SOH prediction than the same current at a moderate temperature. This behavior, learned from the data, directly mirrors the known electrochemical principles that battery engineers use. Therefore, while the mathematical formulations of attention

TABLE III
KEY EXPERIMENTAL SETTINGS AND JUSTIFICATIONS

Hyperparameter/Setting	Value and Justification
Number of FL Clients	22 (NASA), 8 (Oxford), 33 (Tongji). Each battery cell from the respective datasets was treated as a unique client to simulate a non-IID environment. Further details on the datasets are provided in Table II.
Communication Rounds	Up to 1000. An early stopping criterion with a patience of 15 epochs was employed to ensure convergence while preventing overfitting, terminating training when the validation loss ceased to improve.
Data Partitioning	Chronological train/test split. For each client, an 80%/20% split was primarily used. A more aggressive 95%/5% split was employed in scenarios with fewer cycles to maximize the data available for model training.
Sequence Length	Full discharge cycle. The model processes one entire discharge cycle at a time, enabling the Transformer to capture the complete temporal dynamics within a cycle for accurate SOC estimation.
Key Hyperparameters	Tuned via Keras Tuner (Hyperband). Parameters such as the number of Informer-Lite blocks, attention heads, learning rate, and batch size were optimized to minimize validation MAE.
MC Dropout Parameters	Passes (K) = 50, Dropout Rate = Tuned. The number of forward passes (K) was set to 50 for stable uncertainty estimation. The dropout rate itself was optimized during model tuning to balance regularization and uncertainty quality.

and SCAFFOLD are foundational, their application here is specifically tailored. The BFMFA acts as a learned physical proxy, automatically discovering the complex, nonlinear relationships between operational conditions and degradation pathways that are well-established in battery science.

III. SIMULATION RESULTS AND VALIDATION

This section evaluates the proposed TDA-Informer framework trained with the SCAFFOLD federated strategy. We describe the evaluation metrics, experimental setup, and baseline models. Joint SOC and SOH estimation results on the NASA dataset are presented, followed by comparisons with state-of-the-art methods. An ablation study highlights the contribution of key components, and computational efficiency is analyzed to assess suitability for edge applications. Reliability is further examined through UQ and robustness to sensor noise. Generalization is evaluated using the Oxford dataset, demonstrating the framework's adaptability to unseen battery types.

A. Evaluation Metrics and Experimental Setup

To provide a clear and reproducible overview of our experimental design, key settings and their justifications are summarized in Table III. The performance of the proposed and baseline models is evaluated using root-mean-square error (RMSE), MAE, coefficient of determination (R^2), and the standard deviation of per-cycle RMSE (σ_{RMSE}). RMSE and MAE follow standard definitions [52], while R^2 and σ_{RMSE} are defined as

$$R^2 = 1 - \frac{\sum_{\tau=1}^{\mathcal{N}} (\hat{\mathcal{Y}}(\tau) - \mathcal{Y}(\tau))^2}{\sum_{\tau=1}^{\mathcal{N}} (\mathcal{Y}(\tau) - \bar{\mathcal{Y}})^2} \quad (7)$$

$$\sigma_{\text{RMSE}} = \sqrt{\frac{1}{\mathcal{N} - 1} \sum_{\tau=1}^{\mathcal{N}} (\text{RMSE}_{\tau} - \overline{\text{RMSE}})^2}. \quad (8)$$

Here, $\hat{\mathcal{Y}}(\tau)$ and $\mathcal{Y}(\tau)$ denote the predicted and true values at time index τ ; $\bar{\mathcal{Y}}$ is the mean of true values; and RMSE_{τ} is the

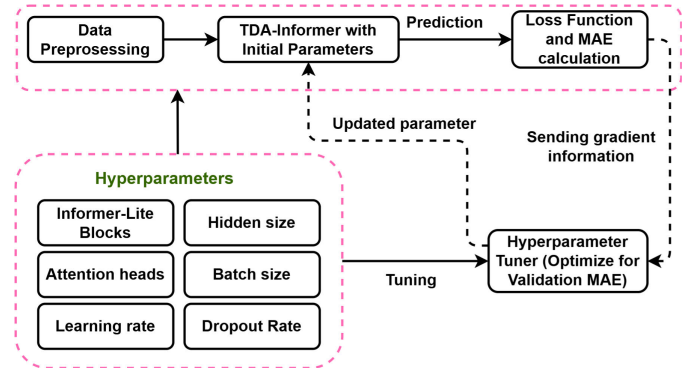


Fig. 4. Hyperparameter tuning workflow.

RMSE for cycle τ . Lower RMSE, MAE, and σ_{RMSE} values indicate better accuracy and consistency, while R^2 closer to 1 implies stronger predictive alignment.

All models were implemented in TensorFlow and trained on a workstation equipped with an Intel Core i9 CPU and NVIDIA RTX 3090 Ti GPU. In the federated setup, each battery was treated as an independent client to emulate non-IID data distributions, with SCAFFOLD used as the default FL algorithm unless stated otherwise. Hyperparameter optimization was performed using Keras Tuner with the Hyperband algorithm, as illustrated in Fig. 4. The search space explored by the Hyperband algorithm and the final optimal values selected for the key hyperparameters are detailed in Table IV. The objective was to minimize validation MAE by tuning the number of Informer-Lite blocks, hidden units, attention heads, learning rate, batch size, and dropout rate. An early stopping criterion with a 15-epoch patience was used to prevent overfitting. Robustness was assessed by injecting 5% Gaussian noise into input features, and UQ was conducted using Monte Carlo Dropout with $K = 50$ stochastic inference passes.

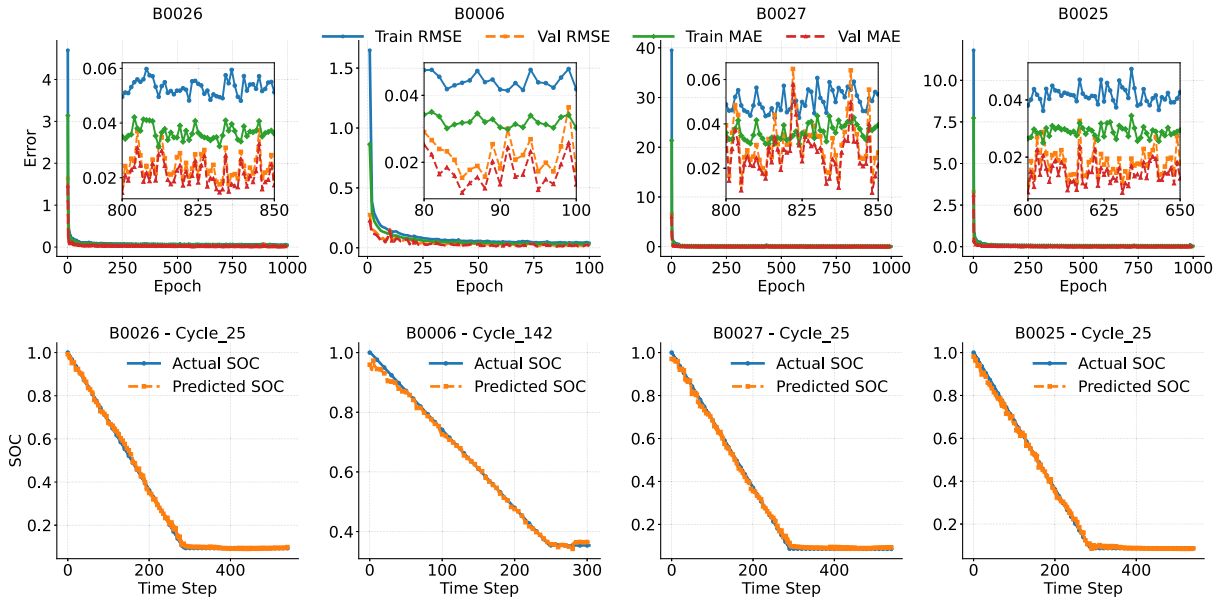


Fig. 5. RMSE/MAE versus epochs (top) and actual versus predicted SOC (bottom) for four NASA battery cells. Insets highlight convergence in later epochs.

TABLE IV
HYPERPARAMETER SEARCH SPACE AND CONFIGURATION

Hyperparameter	Search Space	Final Config.
Informer-Lite Blocks	{1, 2, 3}	Determined individually via Hyperband. Example optimal setting:
Hidden Size (d_{model})	{64, 128, 256}	Blocks=2,
Attention Heads	{2, 4, 8}	Hidden=128,
Learning Rate	{1e-4, 5e-4, 1e-3}	Heads=4, LR=5e-4,
Batch Size	{16, 32, 64}	Batch=32,
Dropout Rate	{0.1, 0.2, 0.3, 0.4, 0.5}	Dropout=0.2

Moreover, we evaluate the effectiveness of the proposed TDA-Informer by comparing it with strong and relevant baseline models, all trained within the same FL framework to ensure a fair and consistent evaluation. The selected baselines include a standard LSTM network [53], representing classical recurrent architectures widely used for battery state estimation; a Vanilla Transformer [54], known for capturing long-range dependencies in time-series data; and the original Informer model [51], which serves as a lightweight Transformer variant optimized for efficiency. These models collectively represent key families of time-series architectures and serve as meaningful benchmarks. This comparison enables a clear assessment of the architectural and algorithmic contributions of TDA-Informer, particularly in terms of multitask learning and feature-focused attention. It is important to note that our validation is conducted in an offline simulated environment. This approach allows for a controlled and reproducible evaluation of the TDA-Informer's accuracy and the SCAFFOLD algorithm's ability to handle data heterogeneity (non-IID). However, this study does not simulate the systems-level network dynamics of a distributed EV fleet. Factors such as communication cost, network latency, and the impact of

asynchronous client updates due to variable connectivity are not captured in the current experiments. These aspects are critical for a practical, real-world deployment and represent a key limitation of the present validation. A full simulation of these network properties is a significant undertaking that we have designated as a primary direction for our future work.

B. Framework Performance Analysis on the NASA Dataset

1) *Joint SOC and SOH Prediction*: The proposed TDA-Informer framework demonstrates robust and reliable learning behavior when evaluated on the NASA battery data, as illustrated across multiple performance visualizations. A total of 22 data, ranging from B0005–B0007, B0018, B0025–B0034, B0036, and B0038–B0044, were used for training and testing purposes. These data span a wide range of ambient temperatures, including 4 °C, 24 °C, and 44 °C, allowing the model to generalize across diverse thermal conditions. In Fig. 5, the training and validation RMSE/MAE curves for four representative cells (B0026, B0006, B0027, and B0025) confirm the model's consistent convergence. The insets in each subfigure further emphasize late-epoch stability, where both training and validation errors flatten with minimal fluctuations, indicating strong generalization without overfitting. Fig. 5 (bottom) also highlights the temporal tracking ability of the model on representative cycles, where the predicted SOC trajectory almost perfectly overlaps with the ground truth. This accurate temporal alignment affirms the model's capability in capturing dynamic battery discharge behavior. Similarly, the SOH predictions for four different NASA cells (B0030, B0029, B0007, and B0018) are shown in Fig. 6. The predictions align closely with the actual degradation curves over the entire lifetime, with both training and testing phases clearly demarcated. The test results maintain fidelity even in later cycles, reinforcing the model's long-term forecasting capability. Further visual support is provided by the boxplots in Figs. 7 and 8. Fig. 7

TABLE V
COMPARISON OF SOC PREDICTION PERFORMANCE FOR THE PROPOSED MODEL AND BASELINE METHODS ACROSS NASA DATA

Dataset	Proposed				Informer				Vanilla Transformer				LSTM			
	RMSE	σ_{RMSE}	MAE	R^2	RMSE	σ_{RMSE}	MAE	R^2	RMSE	σ_{RMSE}	MAE	R^2	RMSE	σ_{RMSE}	MAE	R^2
B0026	0.009397	0.000163	0.006534	0.99899	0.009797	0.000187	0.006714	0.99891	0.012287	0.000208	0.009854	0.99832	0.018811	0.000584	0.015142	0.99618
B0006	0.011466	0.000319	0.007845	0.99705	0.011906	0.000302	0.008655	0.99655	0.016309	0.000535	0.011836	0.99646	0.028583	0.000661	0.025670	0.98960
B0027	0.011470	0.000213	0.008831	0.99854	0.013917	0.000250	0.011811	0.99788	0.016597	0.000275	0.014745	0.99703	0.017089	0.000292	0.015217	0.99685
B0007	0.012749	0.000291	0.009256	0.99712	0.012839	0.000290	0.009362	0.99704	0.015383	0.000349	0.012052	0.99551	0.036410	0.001043	0.032869	0.98276
B0025	0.013033	0.000283	0.009094	0.99810	0.014078	0.000265	0.011860	0.99789	0.014150	0.000272	0.011821	0.99786	0.020008	0.000428	0.017842	0.99586
B0032	0.015616	0.000278	0.013082	0.99639	0.016430	0.000352	0.012868	0.99610	0.016501	0.000329	0.013828	0.99587	0.019195	0.000414	0.016011	0.99498
B0034	0.016262	0.000452	0.011208	0.99587	0.016312	0.000440	0.011454	0.99584	0.031590	0.001437	0.024798	0.98233	0.135374	0.013306	0.116722	0.76769
B0018	0.018954	0.000342	0.016446	0.99306	0.019088	0.000353	0.016597	0.99328	0.021923	0.000512	0.018381	0.99271	0.031652	0.001073	0.027688	0.98673

TABLE VI
COMPARISON OF SOH PREDICTION ACCURACY ACROSS MODELS

Dataset	Proposed			Informer			Vanilla Transformer			LSTM		
	RMSE	σ_{RMSE}	MAE	RMSE	σ_{RMSE}	MAE	RMSE	σ_{RMSE}	MAE	RMSE	σ_{RMSE}	MAE
B0029	0.00204	0.00191	0.00160	0.00806	0.00770	0.00654	0.00798	0.00788	0.00615	0.02223	0.01501	0.01923
B0018	0.00309	0.00300	0.00248	0.00610	0.00592	0.00494	0.00977	0.00926	0.00790	0.02027	0.01086	0.01782
B0030	0.00337	0.00222	0.00278	0.03335	0.01142	0.03133	0.00544	0.00309	0.00470	0.01060	0.00771	0.00841
B0007	0.00541	0.00477	0.00457	0.00941	0.00561	0.00848	0.02037	0.01140	0.01780	0.03059	0.02242	0.02479

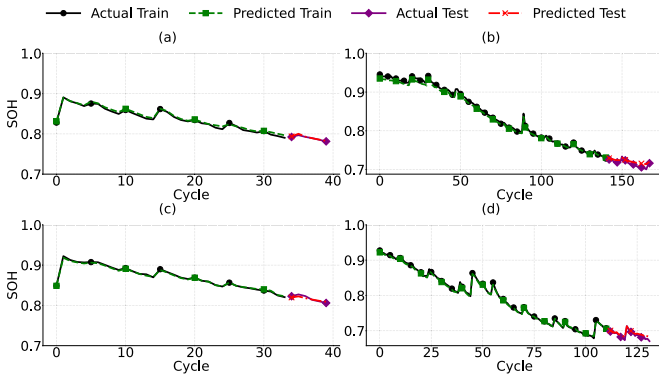


Fig. 6. Comparison of SOH predictions for four NASA battery datasets using the proposed model. (a) B0030, (b) B0029, (c) B0007, and (d) B0018. The figure presents both training and testing results, with clearly distinguished actual and predicted SOH trajectories.

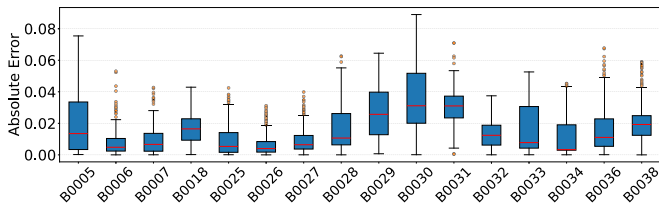


Fig. 7. Boxplot of absolute SOC prediction errors for the best cycle in each dataset. Boxes show IQR, red lines indicate medians, and dots represent outliers.

shows the distribution of absolute SOC prediction errors for the best-performing cycle in each NASA dataset. The boxes exhibit narrow interquartile ranges (IQRs) with most medians below 0.02, confirming prediction robustness across data. In Fig. 8, SOH prediction errors are similarly compact, with minimal outliers and low dispersion across 16 NASA cells. These results collectively validate the proposed method's ability to deliver precise, cycle-level SOC and SOH estimates with minimal degradation in accuracy over time.

2) *Performance Versus Baselines*: Quantitative results across some major NASA cells are summarized in Tables V

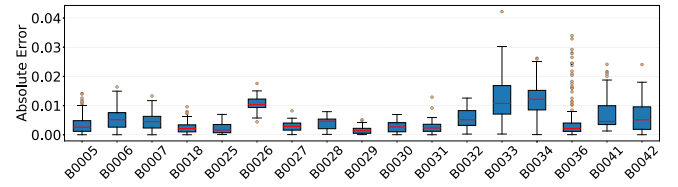


Fig. 8. Boxplot of absolute SOH prediction errors across 16 NASA datasets. Boxes show IQR, red lines indicate medians, and dots represent outliers.

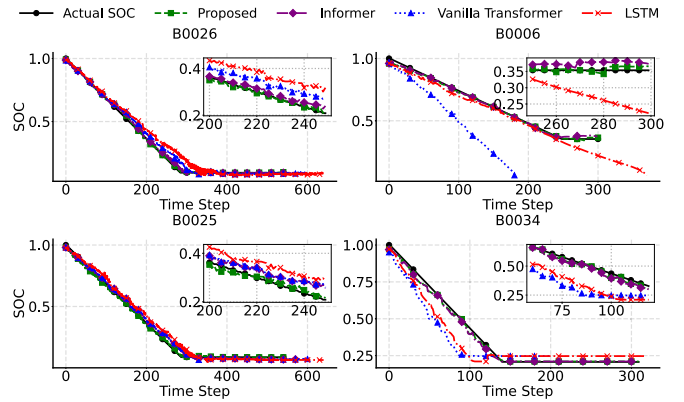


Fig. 9. Comparison of SOC predictions across eight datasets using the proposed model, and three baselines. Insets highlight prediction differences in key segments.

and VI. For SOC estimation (Table V), the proposed model achieves the lowest RMSE and MAE for the majority of the data, with the highest R^2 values consistently exceeding 0.998 in cells such as B0026, B0006, and B0027. Notably, the proposed method maintains a low standard deviation in RMSE (σ_{RMSE}), underscoring its prediction stability. For instance, on dataset B0026, it achieves an RMSE of 0.0094 and σ_{RMSE} of 0.0013, outperforming all baselines, including the Informer and Vanilla Transformer. Similarly, Table VI reflects strong SOH prediction accuracy. The proposed model consistently delivers the lowest RMSE and MAE values across data B0029, B0030, and B0007. For example, on dataset B0029, the

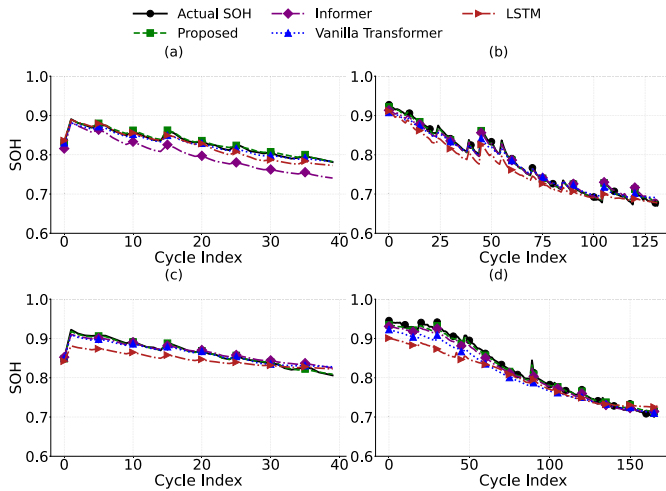


Fig. 10. Comparison of SOH prediction performance for four NASA battery datasets—(a) B0030, (b) B0018, (c) B0029, and (d) B0007—using the proposed model and three baseline methods.

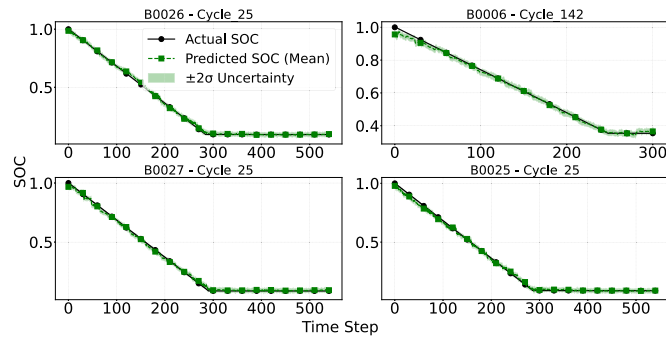


Fig. 11. Predicted SOC with $\pm 2\sigma$ uncertainty for four test cycles from data B0026, B0006, B0027, and B0025.

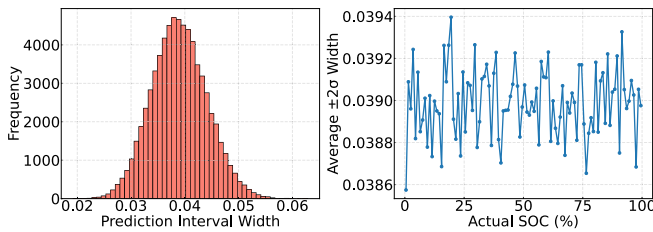


Fig. 12. UQ analysis. (Left) Histogram of $\pm 2\sigma$ interval widths from MC Dropout across all test cycles. (Right) Average $\pm 2\sigma$ width versus actual SOC.

proposed model achieves an RMSE of 0.00204 and an MAE of 0.00160, both outperforming all other compared models. The R^2 scores remain above 0.99 for most cells, highlighting the model's superior learning of degradation dynamics.

Complementing the tabulated metrics, Fig. 9 provides visual comparisons of SOC predictions for selected cells using the proposed model and three baselines. Across all subplots, the TDA-Informer predictions consistently stay closer to the actual SOC curve, especially in regions highlighted by the zoomed insets. Similarly, SOH comparisons in Fig. 10 show that the proposed method captures both the trend and magnitude of degradation more accurately than baseline models, particularly in the test phase. In summary, the consistent outperformance of the proposed model across both error-based and

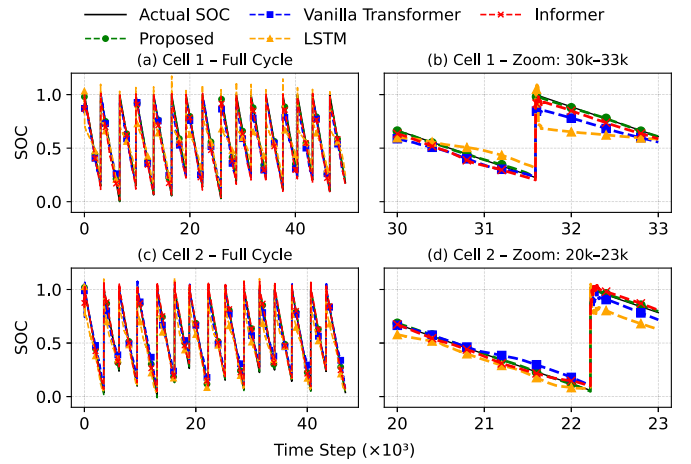


Fig. 13. Comparison of SOC prediction across four models for Cell 1 and Cell 2. (a) and (c) Full cycles and (b) and (d) zoomed-in segments.

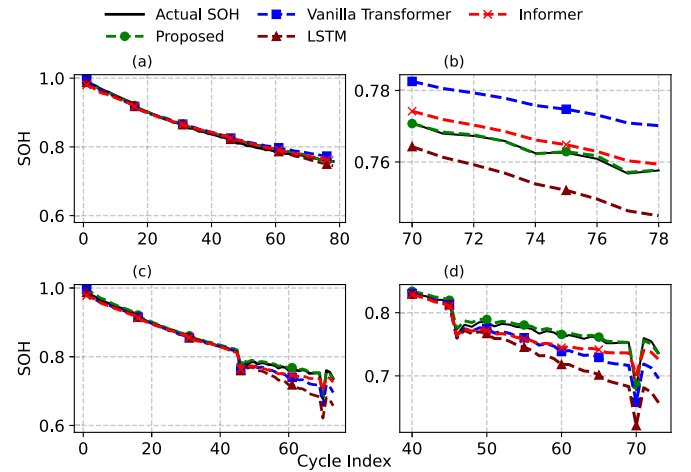


Fig. 14. SOH prediction comparison for Cell 1 and Cell 2 using different models. Each subfigure shows full-cycle and zoomed-in results with adjusted cycle indices. (a) Cell 1 full cycle. (b) Cell 1 zoom: 70–80. (c) Cell 2 full cycle. (d) Cell 2 zoom: 40–80.

regression-based metrics, as well as visual comparisons, validates the effectiveness of the TDA-Informer architecture. The joint modeling of SOC and SOH with shared representations and dedicated task heads, augmented by BFMFA-driven attention modules, facilitates superior generalization, precise tracking, and resilience against temporal and dataset variability.

3) *Analysis of Key Framework Components:* To justify our key design choices and quantify their contribution to the model's performance, we performed two crucial ablation studies on the NASA dataset. The results, summarized in Table VII, confirm that each proposed component provides a significant and measurable improvement. First, we isolated the impact of our advanced FL strategy. By replacing the SCAFFOLD algorithm with conventional FedAvg, the average SOC RMSE increased by 24.8%, and the average SOH RMSE increased by a substantial 38.2%. This result provides strong quantitative evidence that explicitly correcting for client drift with SCAFFOLD is essential for achieving high performance in a realistic, heterogeneous federated environment.

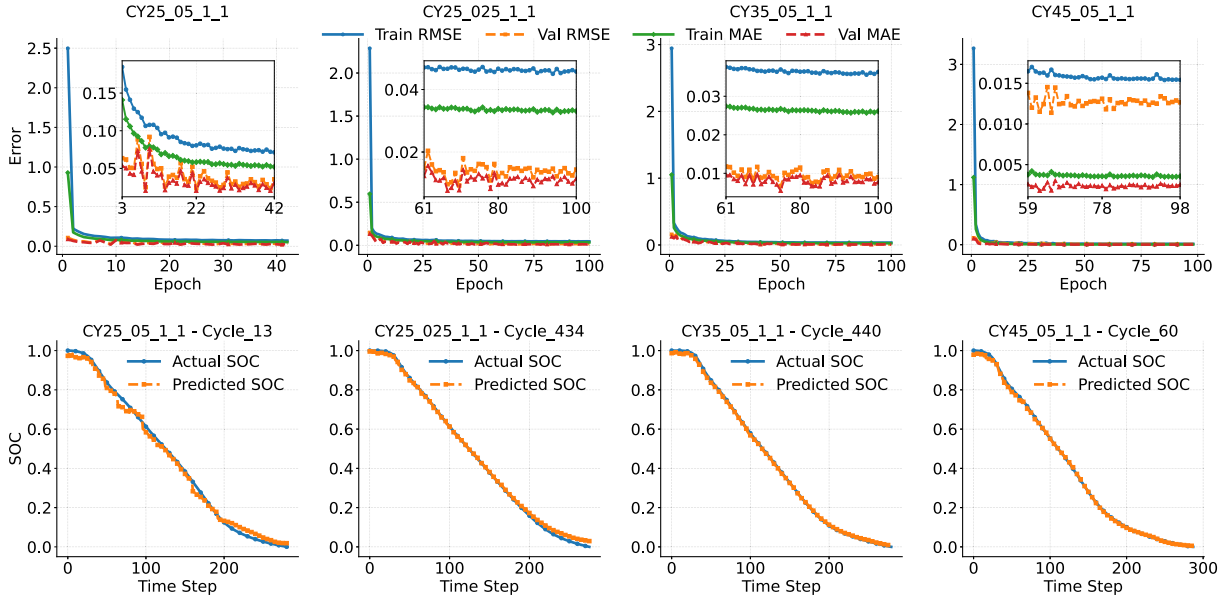


Fig. 15. Training and prediction results for Tongji NCA cells under different operating conditions. The top row illustrates RMSE and MAE variations across epochs, while the bottom row compares actual and predicted SOC profiles for the best test cycles. From left to right, the results correspond to: 25 °C with 0.5C charge and 1C discharge, 25 °C with 0.25C charge and 1C discharge, 35 °C with 0.5C charge and 1C discharge, and 45 °C with 0.5C charge and 1C discharge. Insets highlight convergence behavior during later training epochs.

TABLE VII

ABLATION STUDY RESULTS ON THE NASA DATASET (AVG. RMSE)

Model Configuration	SOC RMSE	SOH RMSE
TDA-Informer + FedAvg	0.0274	0.0102
TDA-Informer without BFMFA	0.0238	0.0083
Proposed Full Model	0.0206	0.0063

TABLE VIII

COMPUTATIONAL PERFORMANCE AND MODEL EFFICIENCY

Model	No. of Parameters	Inference Latency (ms)
LSTM	~0.45 M	5.6
Vanilla Transformer	~2.10 M	14.8
Informer	~1.85 M	12.1
Proposed (TDA-Informer)	~1.20 M	8.7

Second, we evaluated the contribution of our novel attention mechanism by removing the BFMFA module. This change resulted in a 13.4% increase in SOC RMSE and a 24.1% increase in SOH RMSE. This demonstrates that the BFMFA module's ability to dynamically focus on the most salient physical features is a key factor in the model's superior accuracy. Together, these ablation results confirm that both the advanced FL strategy and the feature attention mechanism are essential components that significantly enhance the model's overall performance.

4) *Computational Efficiency for Edge Deployment*: A key objective of this work was to design a model suitable for resource-constrained BMS hardware, a point of concern for many complex architectures. To provide a quantitative validation of our model's efficiency, we analyzed its computational footprint against the baselines, with the results detailed in Table VIII. The proposed TDA-Informer demonstrates a superior balance between predictive power and computational cost. Compared to a Vanilla Transformer, our model reduces the parameter count by a substantial 42.8% (from 2.10 to 1.20 M), which significantly lowers its memory footprint. Crucially, its inference latency of just 8.7 ms is the lowest among all Transformer-based architectures and is well within the real-time processing requirements of a modern BMS. This efficiency gain is a direct result of our lightweight

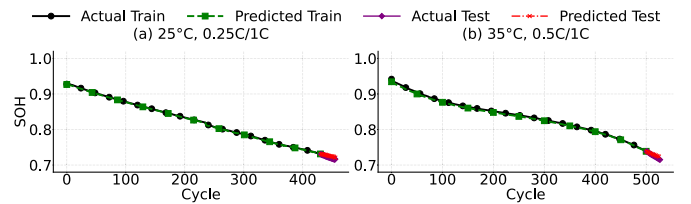


Fig. 16. Comparison of actual and predicted SOH for the NCA Tongji dataset under two different operating conditions. (a) 25 °C, 0.25C/1C and (b) 35 °C, 0.5C/1C.

Informer-Lite design, confirming the practical feasibility of our approach for edge deployment.

5) *Reliability via UQ*: For a safety-critical BMS, quantifying prediction reliability is essential. We integrated a UQ module using MC Dropout ($K = 50$ passes) to estimate the model's epistemic uncertainty. The prediction mean and standard deviation (σ) provide the final estimate and a direct measure of model confidence, respectively. Fig. 11 qualitatively demonstrates this capability, showing that the $\pm 2\sigma$ confidence intervals for four representative cycles are narrow during stable, linear discharge and widen in more volatile, nonlinear regions. This illustrates the model's dynamic, context-aware confidence. This context-aware uncertainty behavior is illustrated in Fig. 12. The histogram

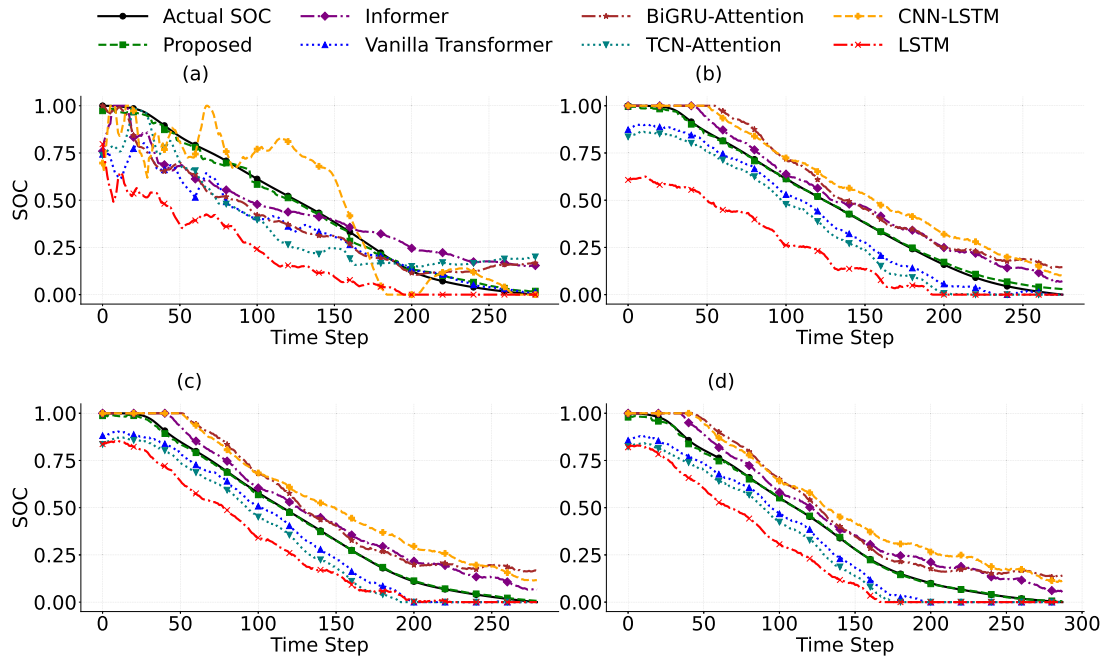


Fig. 17. SOC prediction comparison across four Tongji NCA test cells with representative cycles. (a) CY25_05_1_1-cycle 13. (b) CY25_025_1_1-cycle 434. (c) CY35_05_1_1-cycle 440. (d) CY45_05_1_1-cycle 60.

TABLE IX

ROBUSTNESS TO INJECTED SENSOR NOISE (5% GAUSSIAN NOISE)

Metric (Avg. RMSE)	Clean Test Data	Noisy Test Data
SOC Estimation	0.0206	0.0229
SOH Estimation	0.0063	0.0071

of $\pm 2\sigma$ interval widths shows a tight, unimodal distribution, suggesting that the uncertainty estimates are stable and well-calibrated across test cycles. Additionally, the average interval width tends to increase at low and high SOC ranges (0%–20% and 80%–100%), which aligns with regions typically associated with higher model uncertainty due to non-linear battery dynamics. This behavior supports the potential of the framework to assist in risk-informed decision-making, where predictions in stable SOC regions may be given greater confidence, while caution can be exercised in more uncertain operating zones. Such characteristics contribute positively toward enhancing the practical reliability of the model in battery management contexts.

6) *Robustness to Sensor Noise*: To simulate real-world operating conditions, we tested the resilience of the proposed TDA-Informer model to noisy sensor measurements. Table IX shows that when 5% Gaussian noise was injected into the input features during inference, the model exhibited only a marginal increase in average RMSE, approximately 11.2% for SOC and 12.7% for SOH. Despite this perturbation, prediction accuracy remained well within acceptable limits. This robustness is attributed to the BFMFA-driven attention module, which effectively isolates signal-relevant patterns even in noisy conditions, and the joint multitask learning structure, which enhances the model's overall generalization capacity.

7) *Generalization to Unseen Battery Types*: Robust generalization across battery chemistries and operating conditions is

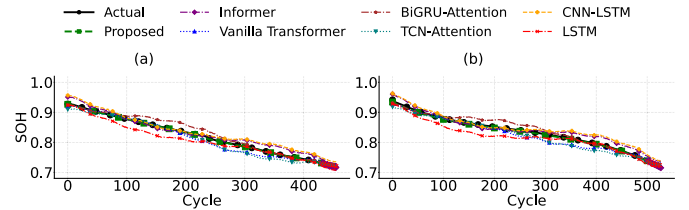


Fig. 18. Comparison of actual SOH, the proposed model, and six comparator models for the NCA Tongji dataset under two operating conditions. (a) 25 °C, 0.25C/1C and (b) 35 °C, 0.5C/1C.

a key criterion for evaluating DL-based battery state estimation models. To assess this capability, the proposed TDA-Informer was trained exclusively on NASA battery data and then directly evaluated on Oxford battery data from eight cells (Cell 1 to Cell 8), all tested under ambient conditions of 40 °C, without any additional training. This dataset presents a substantial domain shift due to differing chemistries (LFP versus LCO), cycle life characteristics, and testing protocols, serving as a rigorous test of model transferability. As shown in Table X, the TDA-Informer outperforms all baselines, including LSTM, Vanilla Transformer, and Informer. For SOC estimation, it achieves the lowest average RMSE (0.0138) and MAE (0.0105), along with the highest R^2 (0.9972) across all eight Oxford cells. For SOH estimation on two representative cells, the model reports an RMSE of 0.0045, MAE of 0.0037, and R^2 of 0.9962, significantly outperforming other methods. These results confirm the model's strong cross-domain predictive capability. Fig. 13 illustrates full-cycle and zoomed SOC predictions for Cells 1 and 2. The TDA-Informer maintains tight alignment with ground-truth across discharge sequences, reducing phase lag and amplitude error compared to baselines. This indicates its ability to learn high-resolution temporal

TABLE X
COMPARATIVE PERFORMANCE ANALYSIS ON THE OXFORD DATASET ACROSS ALL MODELS

Model	SOC Estimation (Avg. over all cells)			SOH Estimation (Avg. over Cells 1–2)		
	RMSE	MAE	R^2	RMSE	MAE	R^2
LSTM	0.1232	0.1002	0.8169	0.0174	0.0110	0.9207
Vanilla Transformer	0.0731	0.0685	0.9671	0.0118	0.0087	0.9723
Informer	0.0257	0.0214	0.9865	0.0084	0.0070	0.9859
Proposed (TDA-Informer)	0.0138	0.0105	0.9972	0.0045	0.0037	0.9962

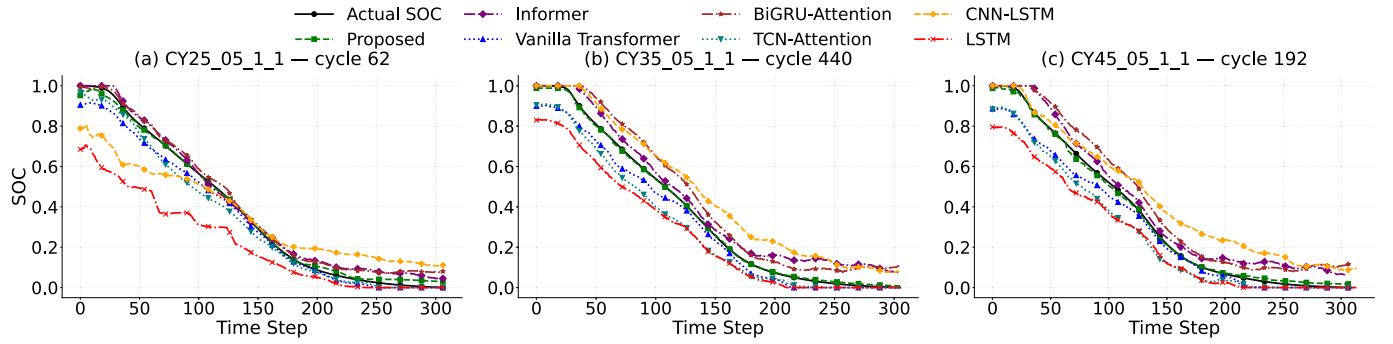


Fig. 19. SOC prediction comparison (NCM) for three cases using representative cycles. (a) CY25_05_1_1, cycle 13, (b) CY35_05_1_1, cycle 440, and (c) CY45_05_1_1, cycle 60.

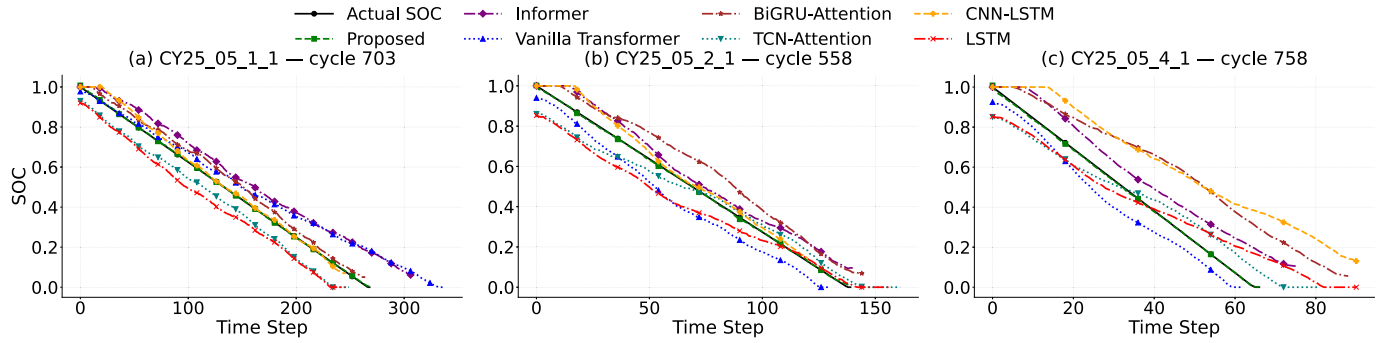


Fig. 20. SOC prediction comparison for Tongji NCA + NCM dataset across three variations using forced or policy-based representative cycles. (a) CY25_05_1_1, cycle 13, (b) CY25_05_2_1, and (c) CY25_05_4_1.

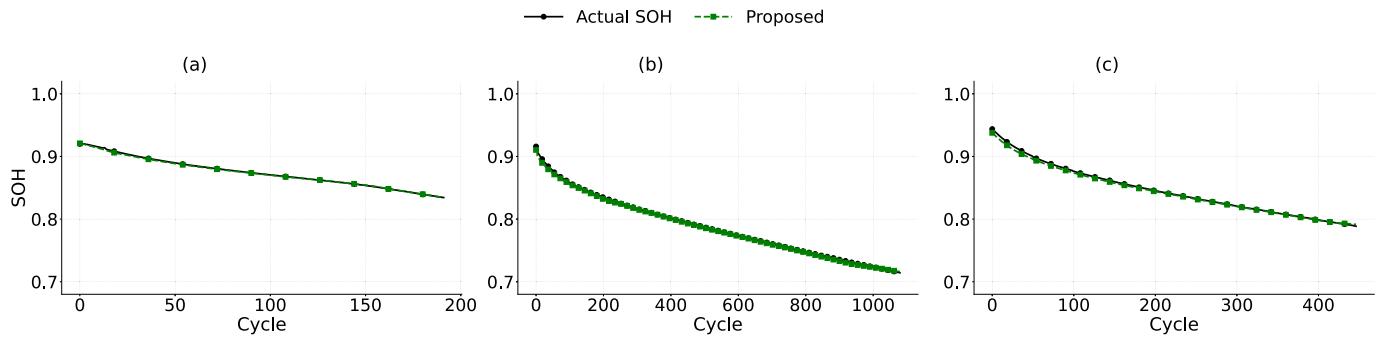


Fig. 21. Comparison of actual and proposed SOH for Tongji NCM dataset under three operating conditions. (a) 25 °C, 0.5C/1C, (b) 35 °C, 0.5C/1C, and (c) 45 °C, 0.5C/1C.

patterns that generalize effectively. SOH prediction trends in Fig. 14 reinforce this conclusion. The proposed model captures both long-term degradation and localized variations near end-of-life, maintaining trajectory fidelity where other models flatten prematurely or show noisy deviations. This robustness is attributed to the joint learning strategy, which

leverages shared degradation patterns across tasks. Overall, the dual-attention BFMFA structure and multitask prediction head enable the TDA-Informer to achieve high accuracy, temporal fidelity, and domain transferability. These qualities establish it as a reliable and scalable solution for real-world battery health monitoring across diverse chemistries.

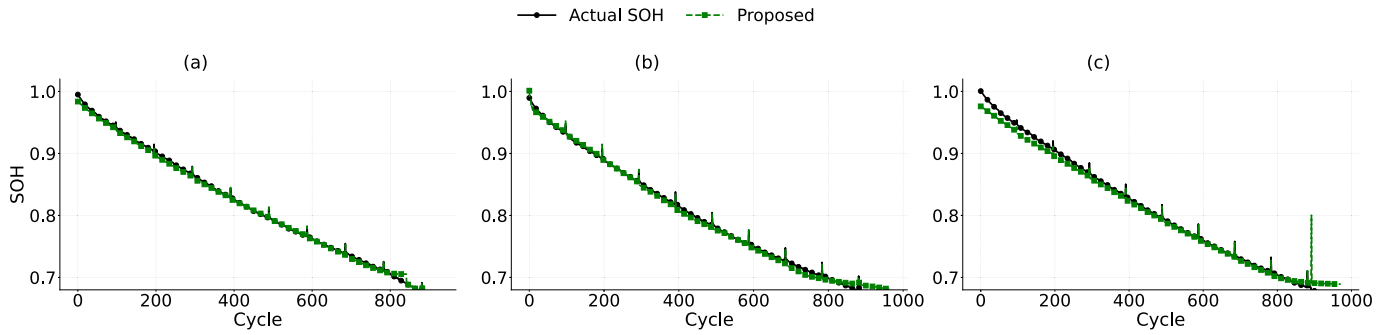


Fig. 22. Comparison of actual and predicted SOH for the proposed model under three NCA+NCM Tongji conditions. (a) CY25_05_1_1, (b) CY25_05_2_1, and (c) CY25_05_4_1.

C. Framework Performance on the Tongji Dataset

To address the limitations of smaller datasets and to evaluate the proposed framework on a modern large-scale dataset with extended life cycles, the Tongji dataset [50] is incorporated. This dataset contains modern commercial chemistries (NCA, NCM, and NCA+NCM) cycled for up to 1500 cycles under diverse operating conditions. To enable a direct comparison of the Transformer-based architecture against other relevant models on long-sequence data, the baseline analysis is expanded to include BiGRU-Attention, TCN-Attention, and CNN-LSTM models. As shown in Figs. 15 and 16, the proposed TDA-Informer model demonstrates strong convergence and accurately tracks both the SOC and SOH degradation trajectories of the Tongji NCA cells used for training, confirming its capability to effectively learn from complex long-sequence datasets. The comparative analysis, presented in Fig. 17 for SOC and Fig. 18 for SOH, indicates that the proposed model consistently outperforms all baseline approaches on the Tongji data. Notably, the achieved accuracy exceeds that of the BiGRU-Attention model, supporting the suitability of the Transformer-based architecture for long-sequence battery data. To further examine generalization capability, the model trained on NASA and Tongji-NCA data is evaluated on the held-out NCM and NCA+NCM chemistries from the Tongji dataset. The results, illustrated in Figs. 19 (SOC comparison on NCM), 20 (SOC comparison on NCA+NCM), 21 (SOH on NCM), and 22 (SOH on NCA+NCM), confirm that the proposed framework maintains high predictive accuracy for these unseen modern chemistries, demonstrating robust cross-domain generalization. The NASA dataset is retained in the study as a widely recognized benchmark to ensure comparability with existing literature.

IV. CONCLUSION

This article presents an FL framework for joint SOC and SOH estimation in EV batteries, addressing the challenges of data privacy and non-IID distributions. We propose TDA-Informer, a lightweight Transformer-based model with a BFMFA module, trained using the SCAFFOLD algorithm. Evaluations on public datasets show improved prediction accuracy over strong baselines and computational efficiency suitable for potential on-device BMS use. The integration of UQ enhances interpretability, while robustness to sensor noise

and generalization to unseen battery types are demonstrated through controlled tests. The proposed federated framework, though validated on single cells, naturally scales to multicell modules and packs. Each cell can be treated as an FL client, with the BMS coordinating training, enabling the global model to capture diverse degradation patterns. Validation on 33 Tongji cells with significant variation demonstrates the framework's ability to handle heterogeneity and extend cell-level estimation toward practical module-level SOH determination, typically governed by the weakest cell. This work offers a practical and scalable foundation for intelligent, privacy-aware battery management. Future work will focus on extending the proposed framework to accommodate various battery chemistries and conducting real-time validation on embedded hardware platforms. In particular, future research will aim to evaluate the framework's accuracy, latency, and computational efficiency under practical deployment conditions. Additionally, an experimental analysis quantifying the performance improvements achieved through local model personalization by comparing the fine-tuned adapter model with the generalized global model will be pursued as a promising direction for future research.

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